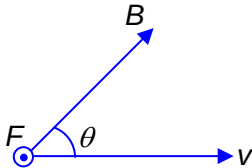
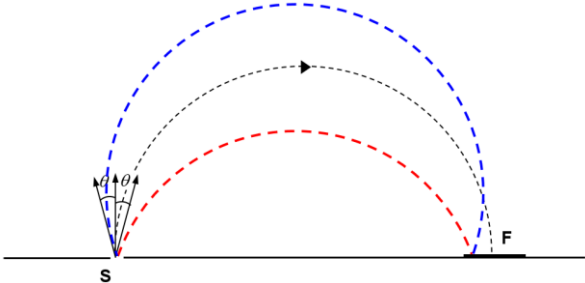


7	(a)	(i)	Moment of a force is the <u>product of the force and the perpendicular distance from the line of action of the force</u> to the point. Torque of a couple is the <u>product of one of the forces and the perpendicular distance between the two forces.</u>		B1 B1
		(ii)	1.	Gravitational field	A1
			2.	Electric field	A1
			3.	Magnetic field	A1
	(b)	(i)	$B = \mu_0 ni.$ $= 4\pi \times 10^{-7} \times 700 \times 3.5$ $= 3.0788 \times 10^{-3}$ $= 3.08 \times 10^{-3} \text{ T}$		A1
		(ii)	$F = BIL$ $= 3.0788 \times 10^{-3} \times 3.5 \times 5.0 \times 10^{-2}$ $= 5.3879 \times 10^{-4}$ $= 5.39 \times 10^{-4} \text{ N}$		M1 A1
		(iii)	$Mgd = F \times \text{length QC}$ $0.10 \times 10^{-3} \times 9.81 \times d = 5.3879 \times 10^{-4} \times 15.0 \times 10^{-2}$ $d = 8.2383 \times 10^{-2} \text{ m}$ $= 8.24 \text{ cm}$		C1 A1
	(c)	(i)			A1
		(ii)	$Bev = \frac{mv^2}{D/2}$ $e/m = \frac{2v}{BD}$		M1 A1
		(iii)	$e/m_e = \frac{2v}{BD}$ $= \frac{2 \times 1.50 \times 10^6}{(2.50 \times 10^{-4}) \times (0.0685)}$ $= 1.75 \times 10^{11} \text{ C kg}^{-1}$		M1 A1

		(iv)	1.	 • Circular shape • Both circles end at LHS of F	A2
			2.	The points of impact on the photographic plate will <u>be nearer than that when θ is zero.</u> This results in the <u>measured value of D to spread</u> over a mean value less than the true value. (The true value of D can be obtained from the largest measured value.)	B1 B1
		(v)	$eE = \frac{mv^2}{D/2}$ $e/m = \frac{2v^2}{ED}$		A1